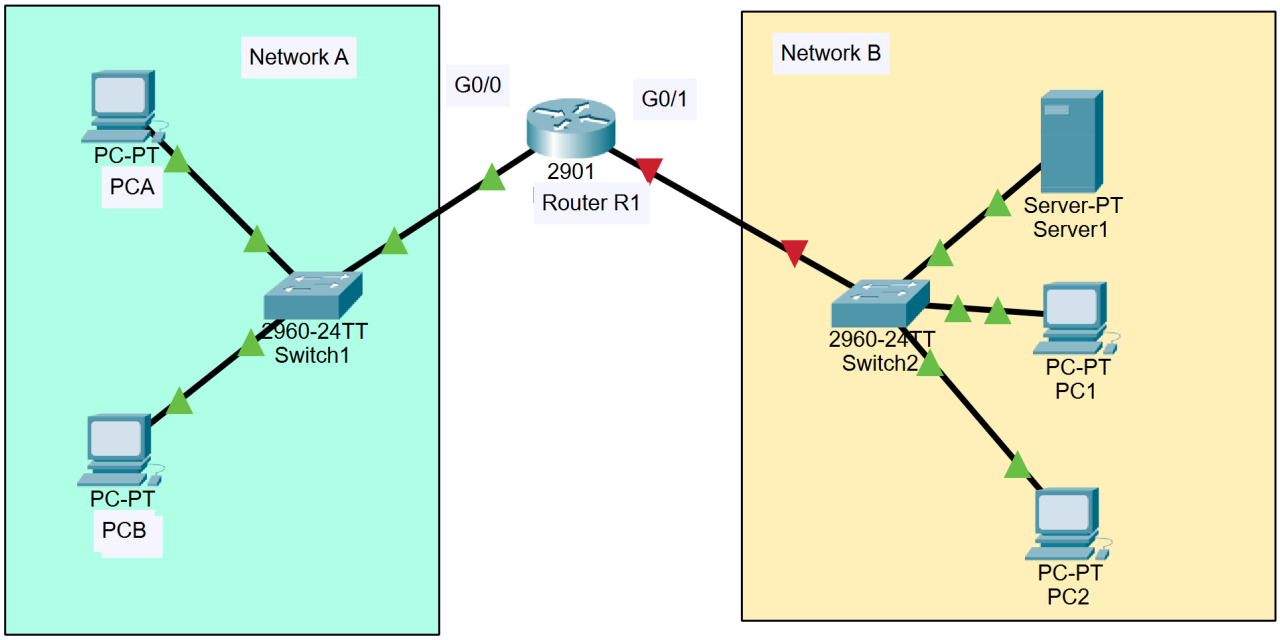
Project 1: Building a Switches and Router Network using CISCO Packet Tracer

Instructions:

1. **This is an individual work.**
2. **Submit on Moodle:**
   a. **A soft copy of your Cisco packet tracer file with your names. (.pkt file) [0.25 Mark]**
   b. **Word document with network design details, all snapshots requested and answer. [2.25 Mark]**

Topology



Step 1: Physical Topology Implementation

Using Cisco packet tracer , Make the network in the diagram above.

Required Resources

- **1 Router (Cisco 2901) – R1: with two LAN interfaces (interfaces GigabitEthernet 0/0 and GigabitEthernet 0/1)**
- **Switches (Cisco 2960)**
- **PCs as end devices**
- **One Web server**

Required Configuration

1) **Cable the network as shown in the topology.**
   a. Layout the devices shown in the topology diagram, and cable, as necessary using cisco packet tracer. Note: you may use auto configuration option for cables.
   b. Power on all the devices in the topology.

2) **Configure the switch.**
   a. No configuration is required on the switch.

3) **Configure the Router.**

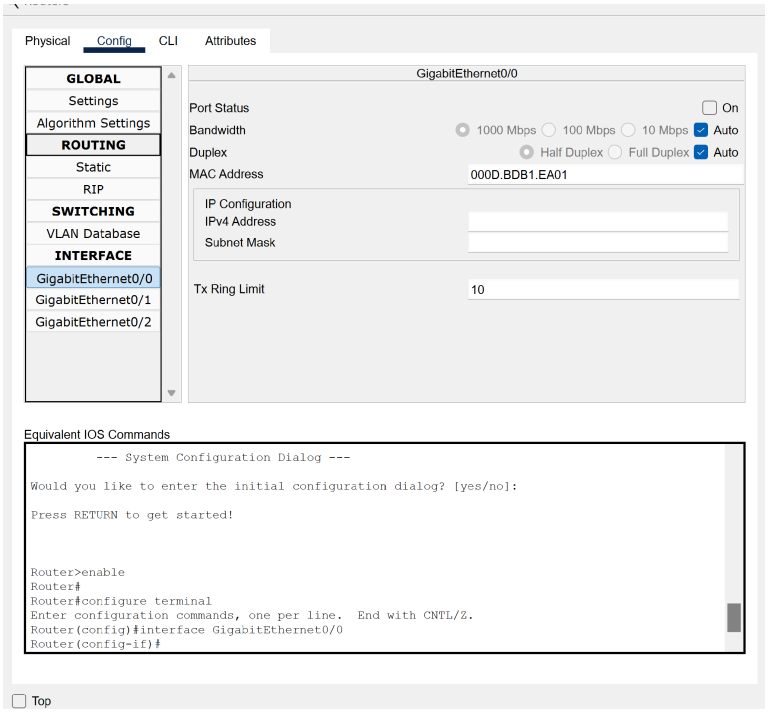
Before configuring the interfaces of the router. You need to design the network and do the subnetting.

A. Network shown above is provided with starting block address of 199.19.30.0/24.

The subnet mask designed for the Company network is /26. Two subnet addresses are used for Network A and Network B. The remaining will be used for future use. Network A is connected to router interface G0/0 and Network B is connected to router G0/1 interface. Fill this table accordingly.

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0			N/A
	G0/1			N/A
PC A	NIC			
PC B	NIC			
PC 1	NIC			
PC 2	NIC			
Web Server	NIC			

B. Configure the router two interfaces GigabitEthernet 0/0 and GigabitEthernet 0/1 with IP addresses and masks according to the table filled. Show the configuration as snapshots.



C. Activate both interfaces up.

Step 4: Configuration on PCs

a. Configure the IP address, subnet mask, and default gateway settings on PC A and PC B.

Show the settings of PC A and PC Bas snapshots.

b. Configure the IP address, subnet mask, and default gateway settings on PC 1, PC 2.

Show the settings of PC 1 and PC 2 as snapshots.

c. Configure the server as **a Web server**. Use the attached document for the configuration.

Show the settings as snapshots.

Step 5: Verifying the network

a. Ping PC A to PC 2. **Put the result as snapshot.**

b. PC B on network A access the webserver on network B. **Put the result as a snapshot.**